

Jupiter's gravity field halfway through the Juno mission

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Contents of this file

Figures S1 to S4

Additional Supporting Information (Files uploaded separately)

Captions for datasets S1 to S2

Introduction

Figure S1 reports the relative position of Juno and the Galilean satellites, to highlight the broad coverage in longitude of Jupiter's tidal bulges.

Figure S2 extends Figure 1 from the main text, reporting all the perijoves.

Figure S3 compares the estimate of low degree gravity coefficients with the solution given by Iess, et al. (2018).

Figure S4 compares the solution obtained with two independent software packages.

Dataset S1 contains the estimated magnitude of empirical accelerations and their uncertainties.

Dataset S2 contains the full covariance matrix for Jupiter's gravity field parameters.

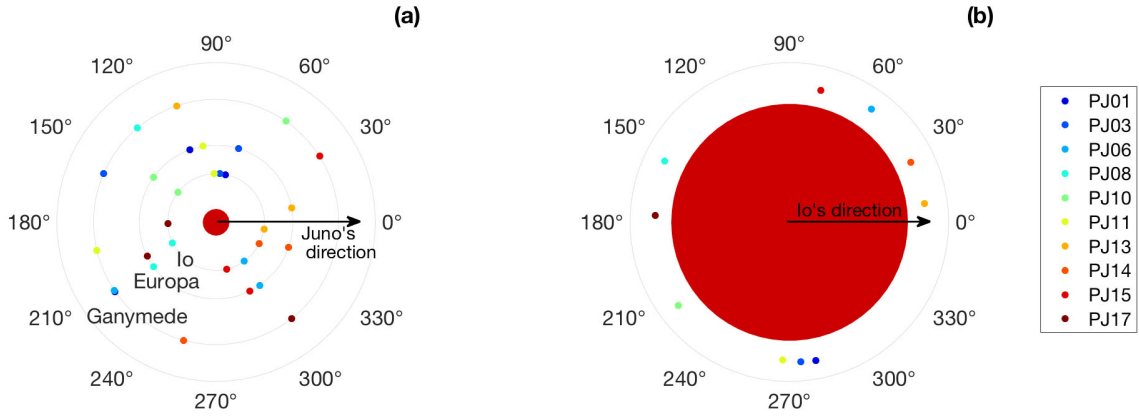
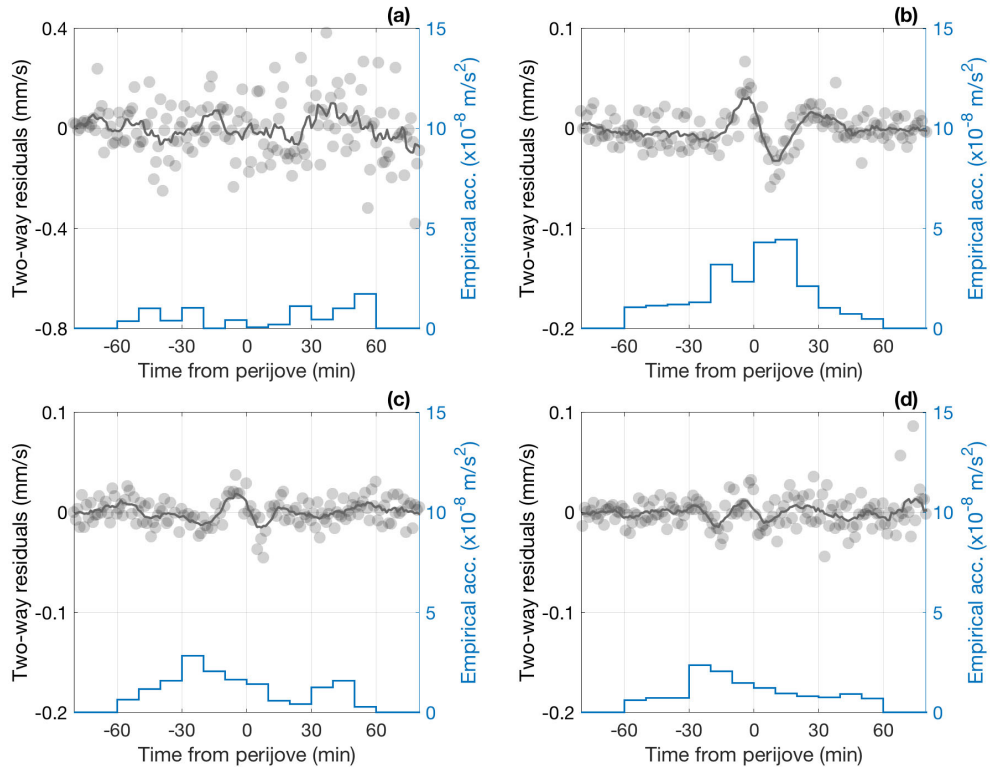


Figure S1. Coverage of tidal bulge. (a) Galilean satellites' angular positions with respect to Juno's perijoves (located at 0°). (b) Juno's angular positions at perijoves with respect to the moon Io (located at 0°).



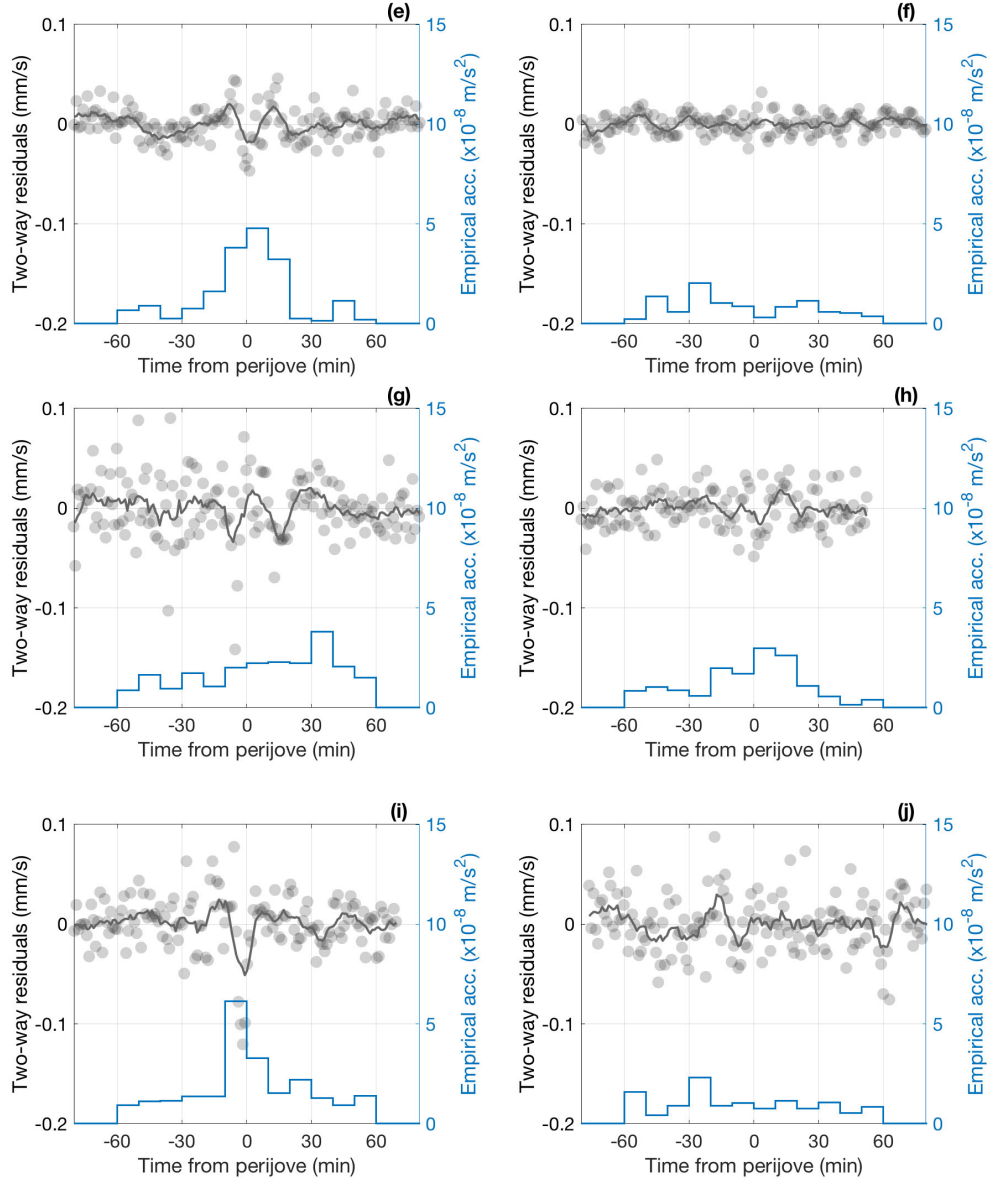


Figure S2. Estimated magnitude of empirical accelerations (blue line) and two-way Doppler residuals (black dots) if we do not include them, for (a) PJ01, (b) PJ03, (c) PJ06, (d) PJ08, (e) PJ10, (f) PJ11, (g) PJ13, (h) PJ14, (i) PJ15, and (j) PJ17. The y-axis limits for PJ01 are different than the rest of the passes given the larger noise. The light black line is a moving average of the residuals highlighting signatures near perijove.

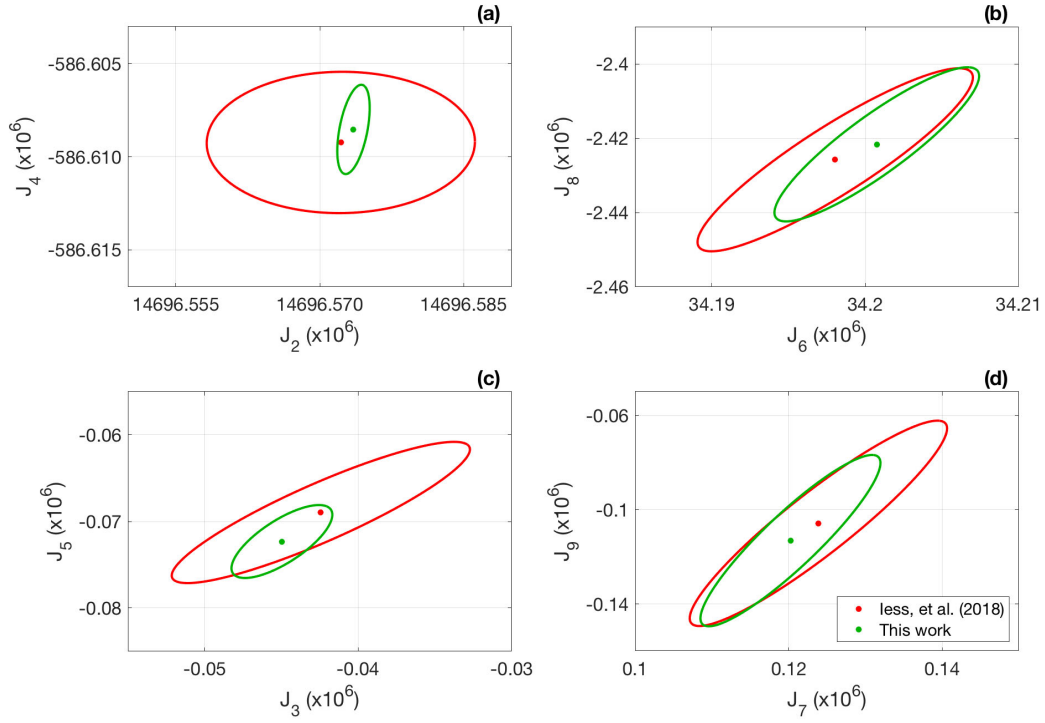


Figure S3. Uncertainty ellipses for pairs of low degree gravity harmonic coefficients. This work's solution (green ellipses) is compared with previous Juno's PJ03 and PJ06 data analysis from less, et al. (2018) (red ellipses). The uncertainties are 3- σ .

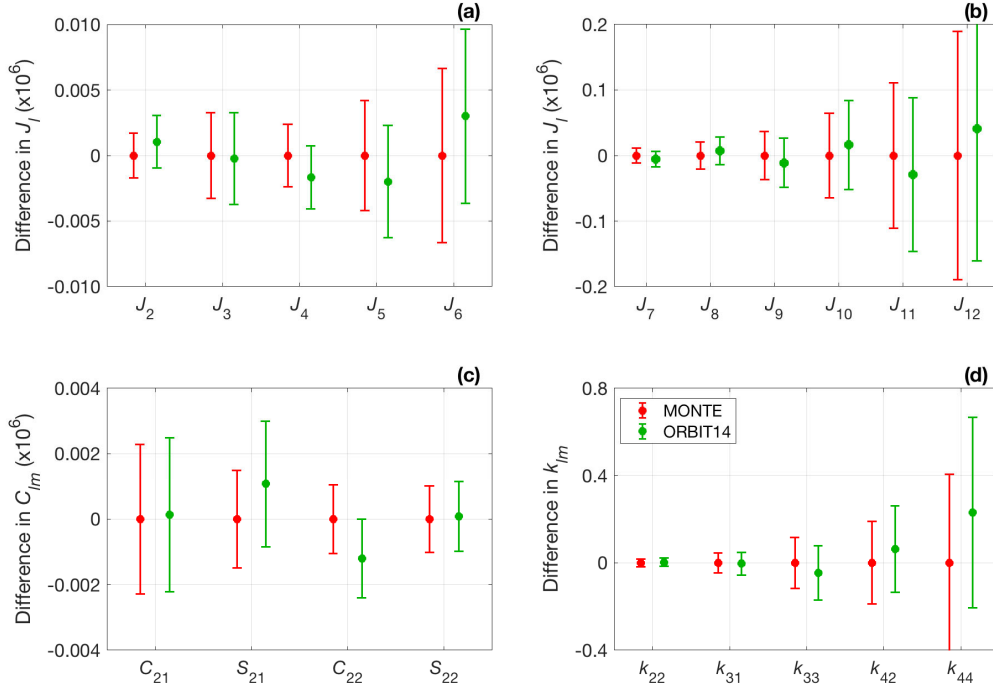


Figure S4. Comparison between the solutions obtained with two independent orbit determination software, JPL's MONTE (reference red values) and University of Pisa's ORBIT14 (green values). The uncertainties are 3- σ .

Data Set S1. Estimated magnitude and uncertainty of empirical accelerations. The columns contains: pericenter label, time from perijove (min), magnitude and uncertainty (m/s^2).

Data Set S2. Covariance matrix for Jupiter's gravity field parameters. The first column is a list of parameter names, which contains: Jupiter's GM (km^3/s^2), fully normalized gravity field coefficients (e.g., $J[2]$), Love numbers (e.g., $k[2][2]$), rotation pole position at J2017.0 (RA and Dec, rad) and rates (rad/s).